

Advanced AC Circuit

Laboratory Exercise No: 5

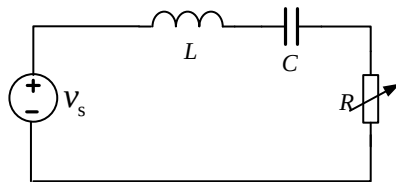
2nd-order RLC Frequency Response

Introduction

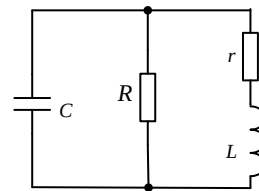
The most prominent feature of the frequency response of a circuit may be the sharp peak (or resonant peak) exhibited in its amplitude characteristic. Resonance occurs in any circuit that has at least one inductor and one capacitor. The simplest resonance circuit is series and parallel resonance. In this lab you will learn how to measure the resonance curve, pass bandwidth, resonance frequency and quality factor in a series resonance and parallel resonance.

1. Series resonance and parallel resonance

- Build the circuit shown in Fig.1 (a) in breadboard, where $L=20\text{mH}$, $C=0.047\mu\text{F}$, $R=10\Omega$, 100Ω .



(a) Series resonance



(b) Parallel resonance

Figure 1. Resonance Circuits

- Use **Function Waveform Generator** as voltage input v_s and use **Digital Oscilloscope** to observe the waveform. Use **Digital Multi-meter** to measure the value of output v_R .

2. Pre-Lab Work

a) Theory analysis

Use phasor analysis method to determine the resonance frequency f_c and quality factor Q for the circuit shown in Fig.1(a). Calculate the voltage over the resistor when $R=10\Omega$, 100Ω , respectively.

b) Simulation Practice

Use Multisim 14.0 to simulate your circuit. Set frequency range from 2kHz-10kHz and make use of **AC Sweep** or employ a **Bode Plotter** to observe frequency response when

$R=10\Omega$, 100Ω , respectively. Use cursor to find the resonance frequency f_0 , cutoff frequencies f_{c1} , f_{c2} and bandwidth B. Output the resonance curve (I - f curve).

c) Use the relevant theory and knowledge and the method of measuring resonance circuit to design a circuit to measure the resonance frequency and resonance curve (V - f curve) for the parallel resonance circuit shown in Fig.1(b) where 'r' is the equivalent DC resistance.

To produce a current source, you can take the circuit shown in Fig. 2 as a reference.

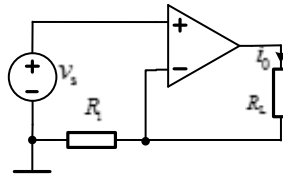


Figure 2

d) use **RLC Meter** to measure R, L and C

(There is only one RLC Meter in the lab. Please come to lab in advance to learn and do the measurement)

In this section you will learn how to use **LCR 6300 Meter** to test an inductor and a capacitor. **LCR 6300 Meter** is used to measure such parameters (under a certain frequency) for an inductor or a capacitor in series and in parallel model as shown in Figure 3 and other parameters such as quality factor Q, dissipation factor D ($D=1/Q$), impedance Z and angle θ_r (in radian) and θ_d (in degree), etc. The relationships of these quantities are

$$Q = \frac{X_S}{R_S}, \quad D = \frac{1}{Q} = \frac{R_S}{X_S}, \quad X_S = 2\pi f L_S = \frac{1}{2\pi f C_S}$$

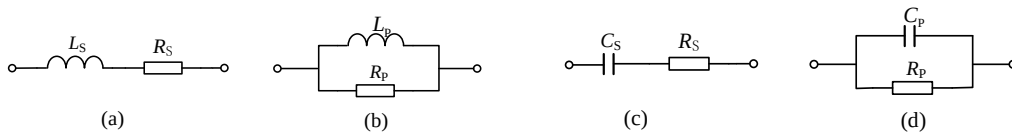


Figure 3

The front panel of the instrument is shown in Figure 4. A brief introduction is as following

- | | | |
|---------------------|-------------------|----------------------|
| 1 — LCD display; | 2 — soft keys; | 3 — number keys; |
| 4 — direction keys; | 5 — power switch; | 6 — measure key; |
| 7 — setup key; | 8 — input ports; | 9 — trigger key; |
| 10 — ESC key; | 11 — USB host; | 12 — system function |

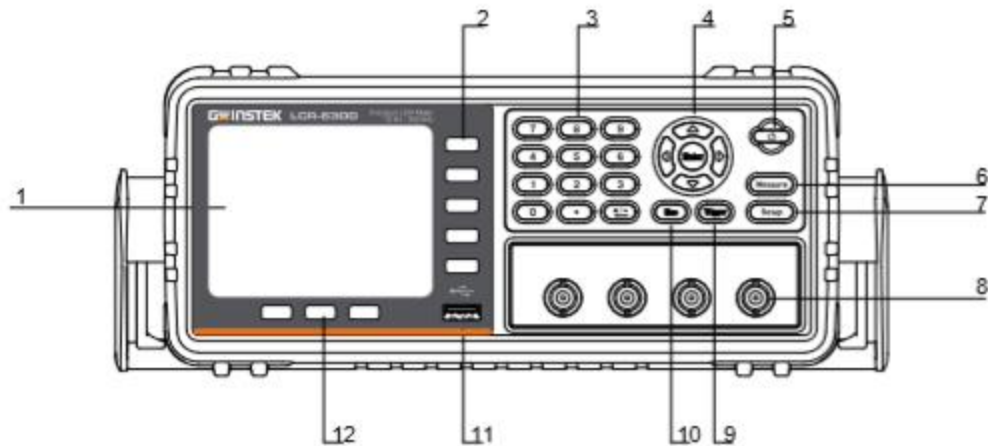


Figure 4

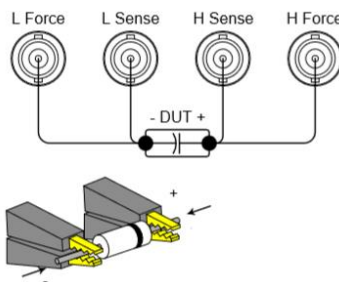


Figure 5

Steps to test R, L or C

- (i) Switch on the **LCR 6300 Meter** and start to measure after 30 minutes (the measurement will be more accurate);
- (ii) Press **Setup** key and press **Direction** keys to select parameter combinations;
- (iii) Press **Measure** key to display (use **direction** keys to select **FUNC** or **FREQ**. Use **number** keys to input the value of frequency and then **Enter**);
- (iv) Connect DCT (device under test) as shown in Figure 5;
- (v) Read the result or save the measurement (**Measure** → LOG → START LOG → SAVE TO USB).

4. Lab Section 1

In this laboratory, you will learn how to measure the RLC resonant frequency, quality factor, resonant characteristic curve and bandwidth. Build the circuit shown in Fig.1 on a bread board. Use **Function Generator** as input. Set $v_{s\text{ P-P}}$ to 1 V and set the frequency from 2kHz to 10kHz. Use **Digital Multi-meter** to measure voltage and use **Oscilloscope** to monitor voltage and phase. The procedure includes the following steps:

(i) **Element measurement.** Use **LCR 6300 Meter** to measure the resistance rated $R=10\ \Omega$, $100\ \Omega$, capacitance rated $C=47\ \text{nF}$ (C_s , R_s and Q_c) and inductance rated $22\ \text{mH}$ (L_s , R_s and Q_L)(at resonant frequency or nearby).

(ii) **Circuit building.** Build the circuit shown in Fig.1(a) on breadboard($R=10\ \Omega$). Select Sine Function from **Function Waveform Generator** as v_s , and monitor it on the **Digital Oscilloscope** (CH1). Use CH2 to monitor current(v_R/R). Use **Digital Multi-meter** to measure voltage.

(iii) **Resonance frequency measurement.** Change frequency from $2\ \text{kHz}\sim 10\ \text{kHz}$. Observe the output voltage, when it reaches maximum, the corresponding frequency is the resonance frequency. Substitute the $10\ \Omega$ to $100\ \Omega$ to see if the resonance frequency changes.

Another way to find resonance is to set oscilloscope to work in X-Y mode. The phase difference will show an ellipse on oscilloscope. Change the frequency, when the ellipse changes to a line, the circuit is resonant.

Measure the voltage over the capacitor at the resonance frequency you have got. Change the frequency to see if it is maximum value at this frequency.

(iv) **Resonance curve measurement.** Keep the input voltage all the same. Set frequency from $2\ \text{kHz}\sim 10\ \text{kHz}$. Measure the resonance curve (I - f curve) when $R=10\ \Omega$ and $100\ \Omega$, respectively. Mark the cutoff frequency on lower side and higher side on the curve. Calculate the bandwidth.

(v) Compare the results with those from simulation.

Requirements and notes

(i) While measuring, every time you change the frequency, make sure to adjust v_s so that it keeps unchanged.

(ii) Use either double traces method or ellipse method to find resonance frequency.

(iii) Fill all the results in the data table you made yourself.

(iv) Sketch I - f curve and calculate quality factor $Q_0 = f_0 / B_f$. Is there any relationship among Q_0 , Q_L , and Q_C ?

(v) Calculate V_{C0}/V_s to see if it is Q_0 .

(vi) Compare the lab data with theoretical results. Analyze the reason that may produce the error.

5. Lab Section 2

In this laboratory, you will learn how to measure a RLC resonance in parallel. The procedure includes the following steps:

- (a) Make use of what you have learned to design an RLC parallel circuit for measurement shown in Figure1(b).
- (b) Measure the resonance frequency and v - f curve.
- (c) Assess your design by theory analysis along with simulation or practical test.

6. Submission

- a) Complete the theory analysis;
- b) Complete the software simulation including the simulating circuits, amplitude and phase response plots, measurement results;
- c) Complete the practical measurements including data filling in forms and I - f curve;
- d) Complete the analysis as required in Lab Section1.
- e) Make sure the names and IDs of **BOTH** team members are on the report,
- f) You should submit the completed report to the Moodle before due time

7. Assessment

- i) This lab is assessed by 100marks.
- ii) The prelab work is assessed together with the lab work.
- iii) The lab work is designed to be completed in groups of two. Each group is required to submit ONE report which is assessed by 100%.